MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE DBF

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.4 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DBF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/157-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current and output impedance apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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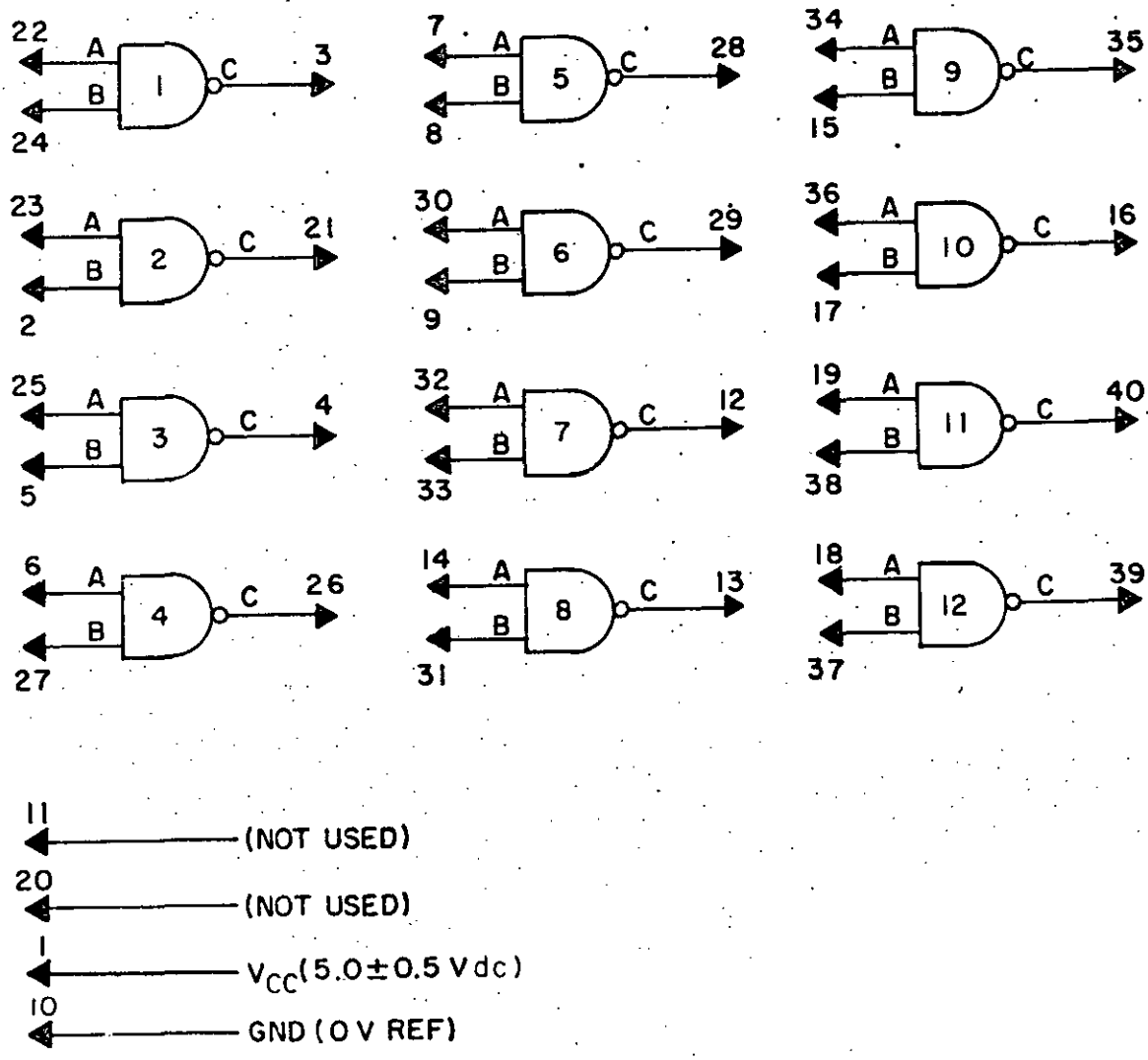


FIGURE 1. DBF module function diagram, NAND gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B24	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	μA
C1	Isolation, frame	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
C2-C3	Isolation, unused pins	V	X	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D12	Output voltage, low level	VI	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
E1-E24	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F24	Input current, low level	VIII	X	50	200	50	200	50	200	50	200	μA
G1-G24	Input current, high level	IX	X	0	10	0	10	0	10	0	10	μA
H1-H12	Output short circuit current	X	X	3	15	3	15	3	15	3	15	mA
J1	Power supply current, low level	XI	S	0.5	6.1	0.5	6.1	0.5	6.1	0.5	6.1	mA
J2	Power supply current, high level	XI	S	0.2	2.4	0.2	2.4	0.2	2.4	0.2	2.4	mA
K1-K12	Transition time (t_{TLH}), C_L = 40 pF	XII	S	5	60	5	60	5	60	5	60	ns
K13-K24	Transition time (t_{TLH}), C_L = 140 pF	XII	Q	10	100	10	100	10	100	10	100	ns
L1-L12	Transition time (t_{THL}), C_L = 40 pF	XIII	S	2	30	2	30	2	30	2	30	ns
L13-L24	Transition time (t_{THL}), C_L = 140 pF	XIII	Q	5	60	5	60	5	60	5	60	ns
M1-M12	Propagation delay time (t_{PLH}), C_L = 40 pF	XIV	S	10	60	10	60	10	70	10	70	ns
M13-M24	Propagation delay time (t_{PLH}), C_L = 140 pF	XIV	Q	10	90	10	90	10	100	10	100	ns
N1-N12	Propagation delay time (t_{PHL}), C_L = 40 pF	XV	S	10	82	10	82	10	85	10	85	ns
N13-N24	Propagation delay time (t_{PHL}), C_L = 140 pF	XV	Q	10	96	10	96	10	100	10	100	ns
P1-P12	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
R1-R12	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVIII	Q	0	300	0	300	0	300	0	300	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T	Power supply filter capacitance	XIX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
U1-U24	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF
V1-V24	Output impedance, power off, check	XXI	Q	3.50	5.55	3.50	5.55	3.50	5.55	3.50	5.55	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/157-1 supersedes part number M28787/157. The two parts are interchangeable.

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

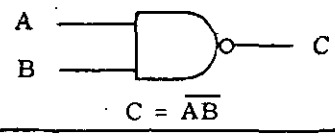
Maximum power dissipation	
All outputs at low level	34 mW (outputs not loaded)
All outputs at high level	14 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current	
Low level (V_{IL} = 0.0 V)	-0.2 mA maximum
(V_{IL} = 0.3 V)	-0.18 mA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	100 µA maximum
Output voltage	
Low level (I_{OL} = 2.0 mA)	0.3 V maximum
High level (I_{OH} = -100 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.3 V)	2.0 mA
High level (V_{OH} = 2.4 V)	-100 µA

6.5 Module function. The module contains 12 independent low power 2-input NAND gates.

6.5.1 *Input to output relationship.* Each NAND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Input		Output
A	B	C
L	L	H
L	H	H
H	L	H
H	H	L



$C = \overline{AB}$

H = High level
L = Low level

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6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

The circuit groups which can be paralleled are:

a. Circuits 1, 2, 3, and 4.

b. Circuits 5, 6, 7, and 8.

c. Circuits 9, 10, 11, and 12.

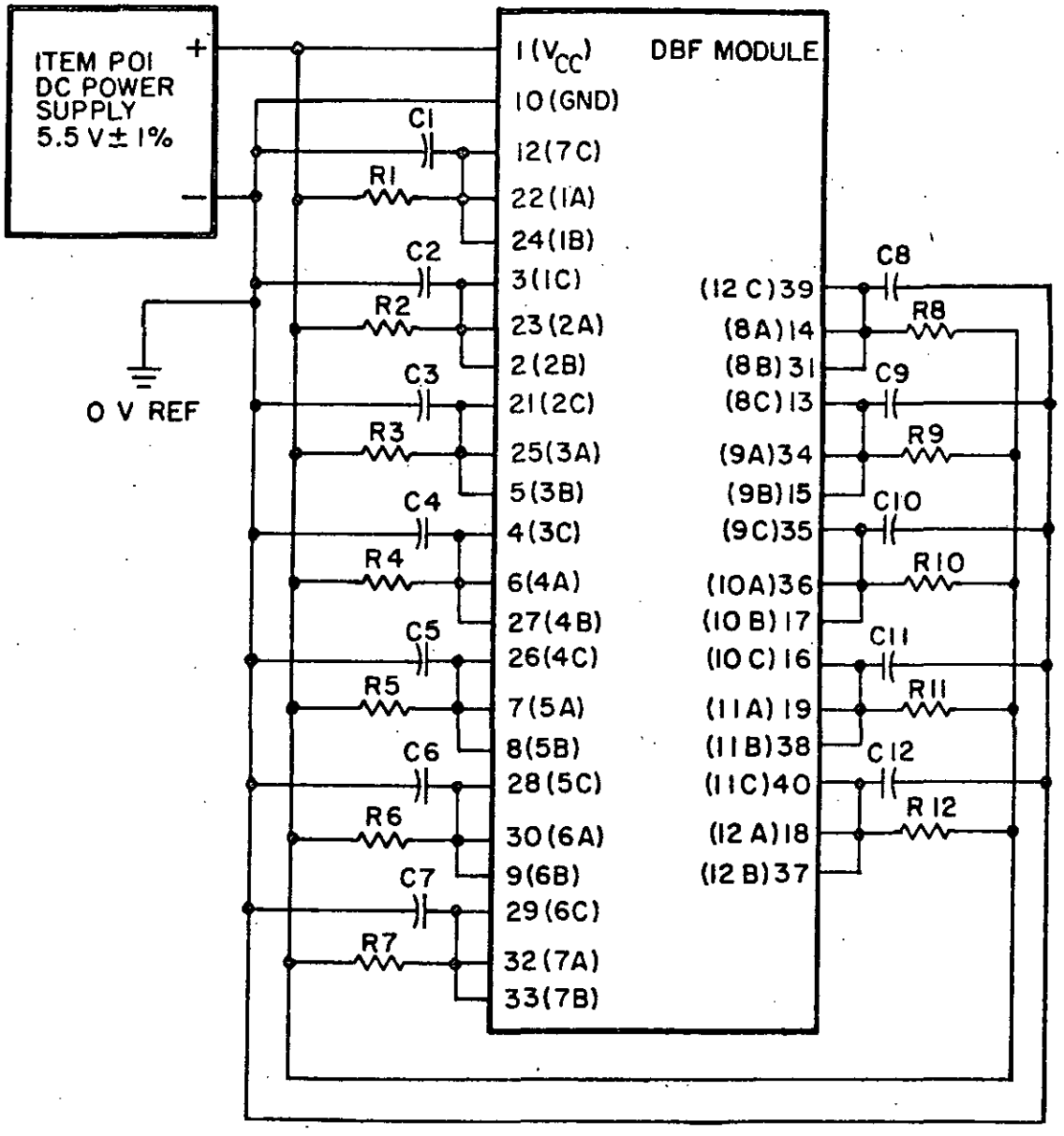
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
DLA - ES

Preparing activity:
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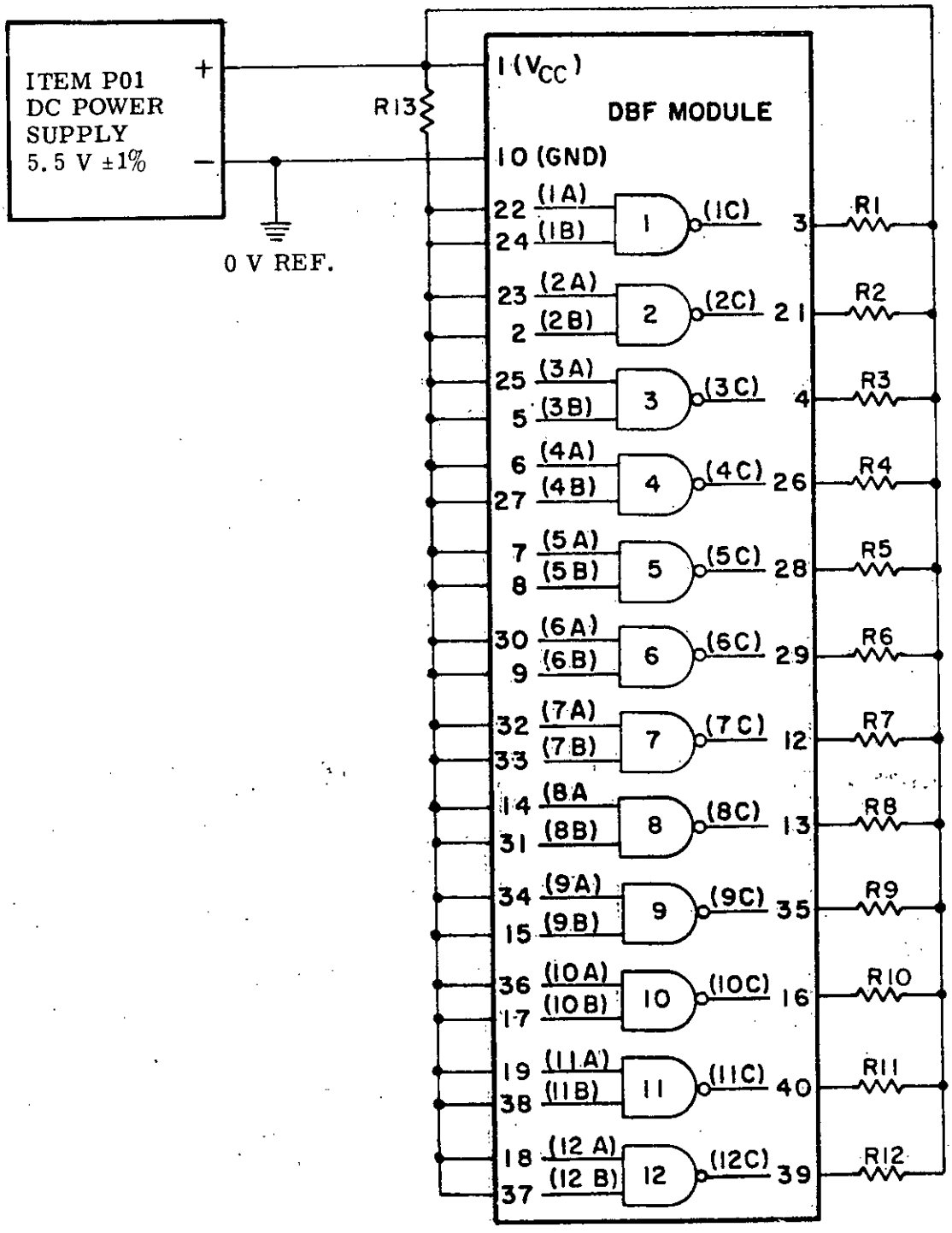
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NOTES:
1. R1-R12 = 2.75 kΩ ±1%, 1/8 W.
2. C1-C12 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R13 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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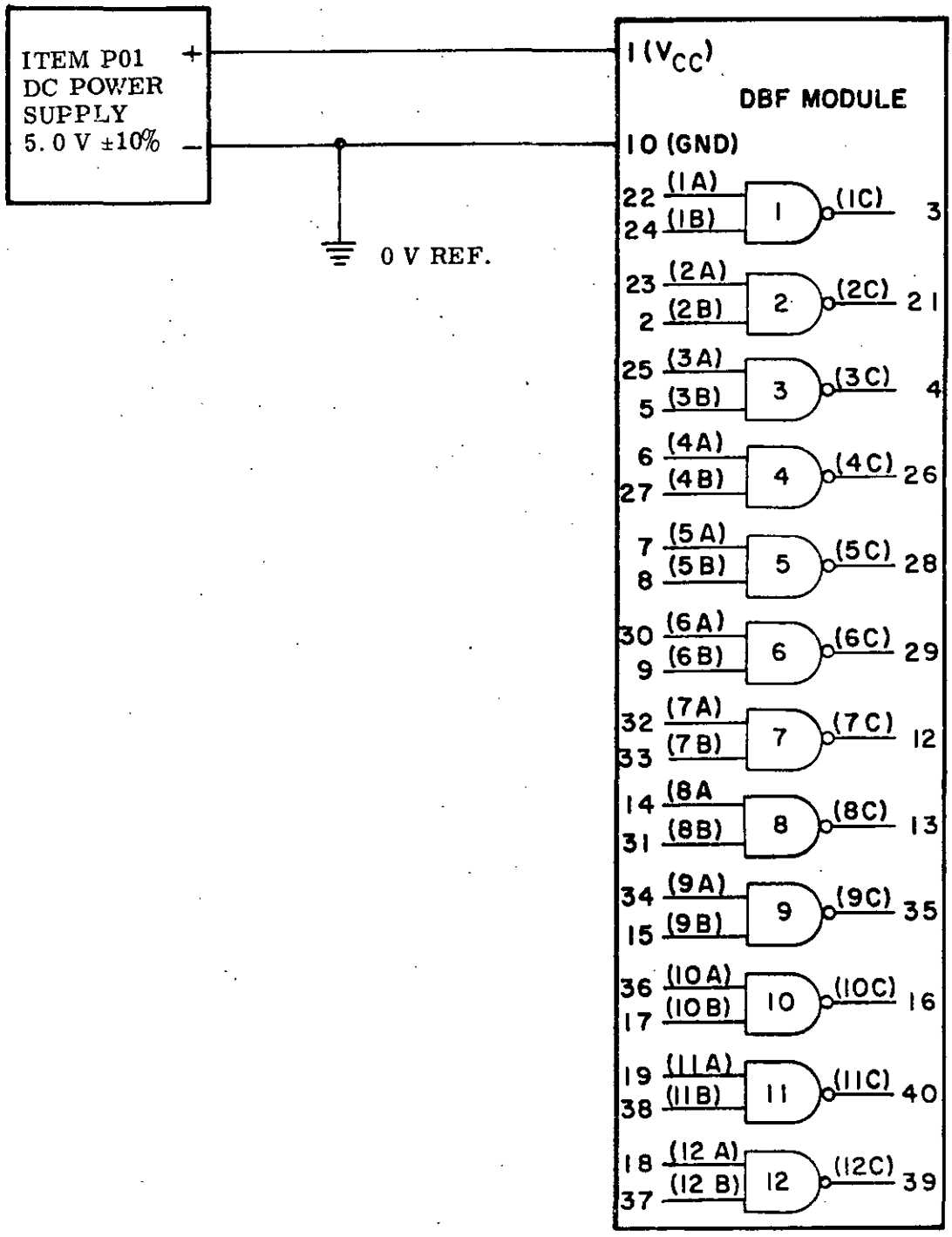


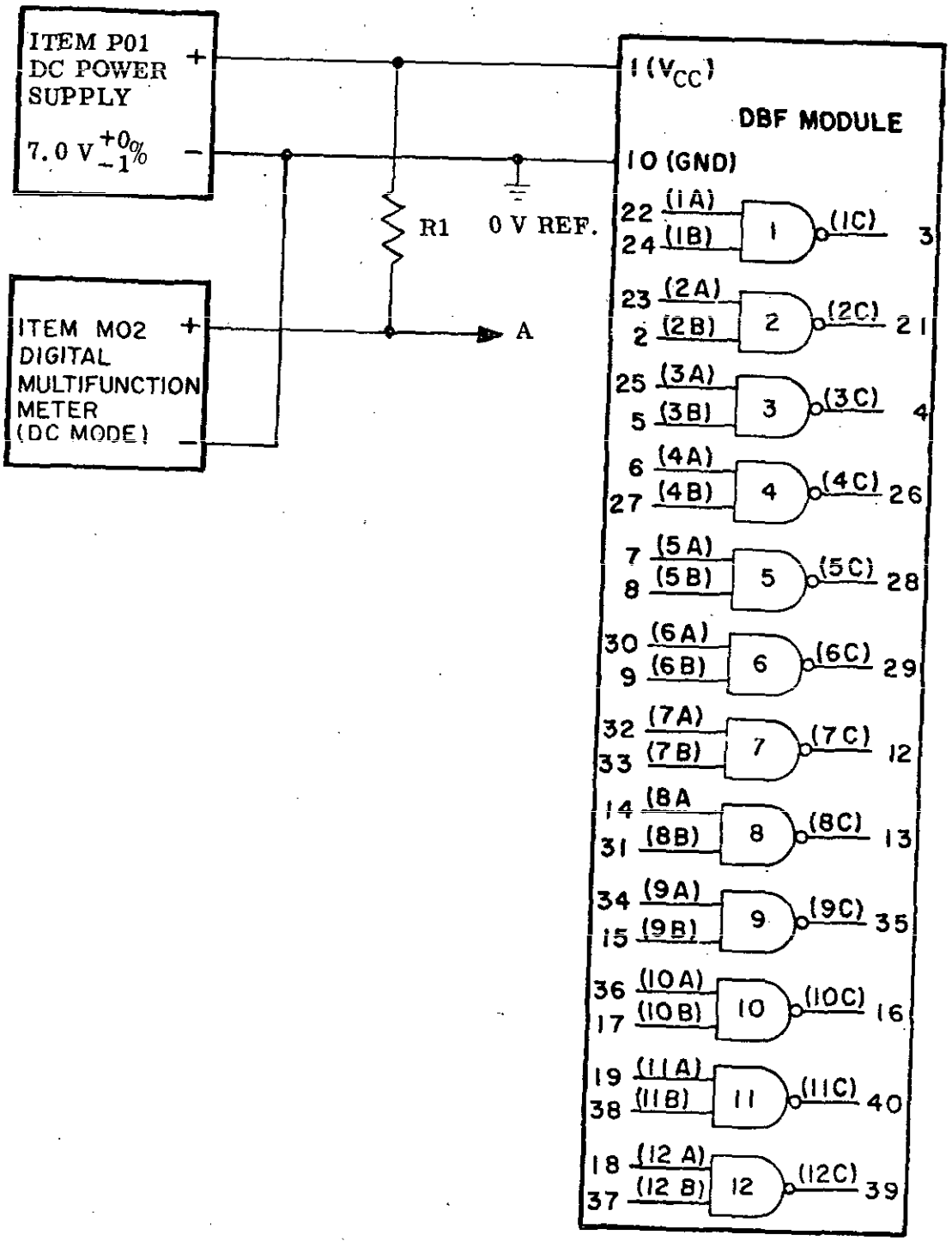
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1C V_{OUT} (low loaded)	3	0.0	0.6	V	5
A2	2C	21	↓	↓	↓	↓
A3	3C	4				
A4	4C	26				
A5	5C	28				
A6	6C	29				
A7	7C	12				
A8	8C	13				
A9	9C	35				
A10	10C	16				
A11	11C	40				
A12	12C	39				

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NOTES:
1. R1 = 2.75 kΩ ±1%, 1/8 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

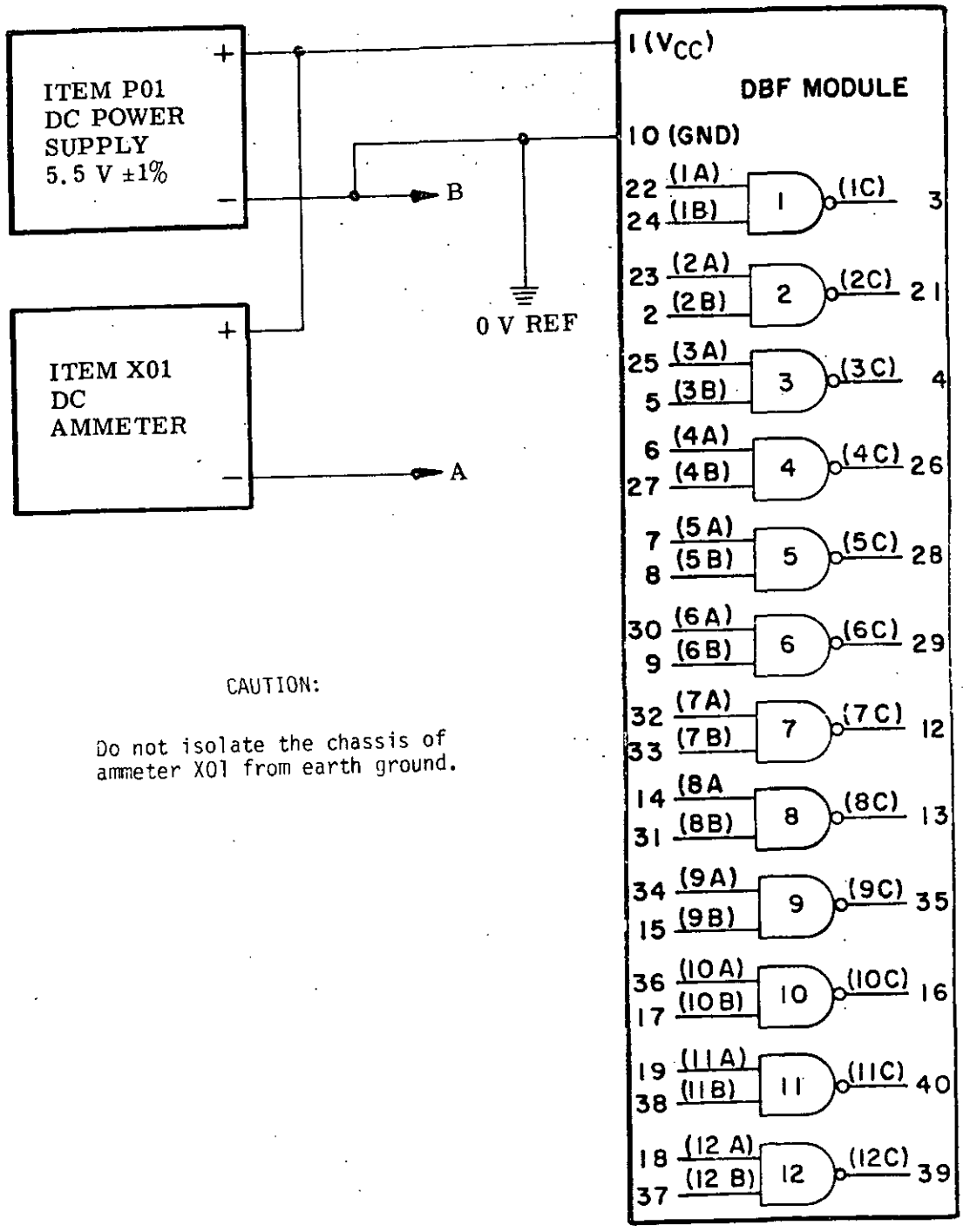
FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input breakdown current	22	24	0	100	µA	6
B2	1B	24	22				
B3	2A	2	23				
B4	2B	23	2				
B5	3A	5	25				
B6	3B	25	5				
B7	4A	6	27				
B8	4B	27	6				
B9	5A	7	8				
B10	5B	8	7				
B11	6A	9	30				
B12	6B	30	9				
B13	7A	32	33				
B14	7B	33	32				
B15	8A	14	31				
B16	8B	31	14				
B17	9A	15	34				
B18	9B	34	15				
B19	10A	17	36				
B20	10B	36	17				
B21	11A	19	38				
B22	11B	38	19				
B23	12A	18	37				
B24	12B	37	18				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

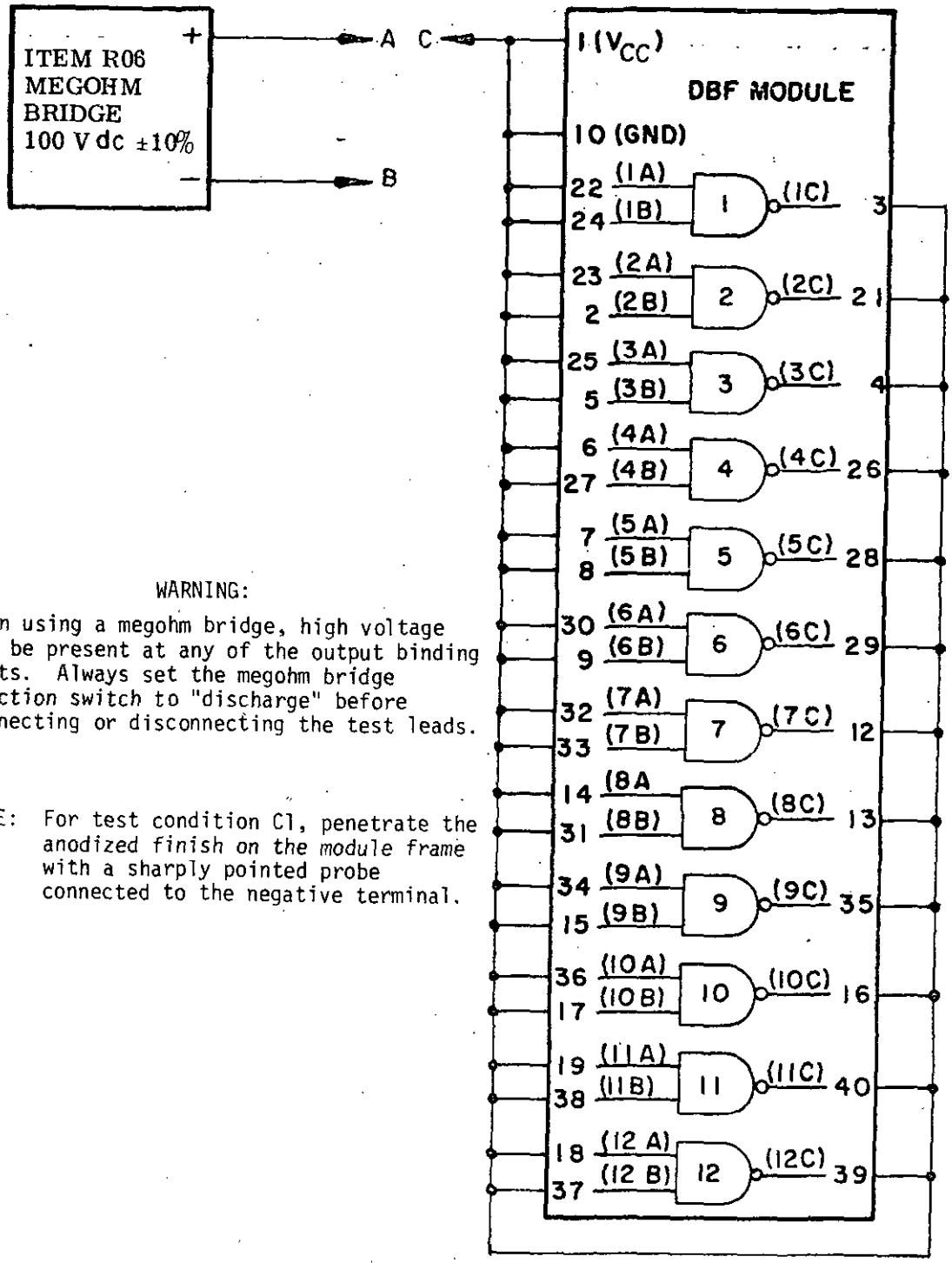
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, frame to module pins	C, 11, 20	Frame	10	∞	MΩ	7
C2	Isolation, used to unused pins	C	11, 20	10	∞	MΩ	7
C3	Isolation, unused to unused pins	11	20	10	∞	MΩ	7

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition C1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1C Output voltage, low level	22, 24	Open	3	0.0	0.3	V	8
D2	2C	2, 23		21				
D3	3C	5, 25		4				
D4	4C	6, 27		26				
D5	5C	7, 8		28				
D6	6C	9, 30		29				
D7	7C	32, 33		12				
D8	8C	14, 31		13				
D9	9C	15, 34		35				
D10	10C	17,36		16				
D11	11C	19, 38		40				
D12	12C	18, 37		39				

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TABLE VII. Output voltage (high-level) test.

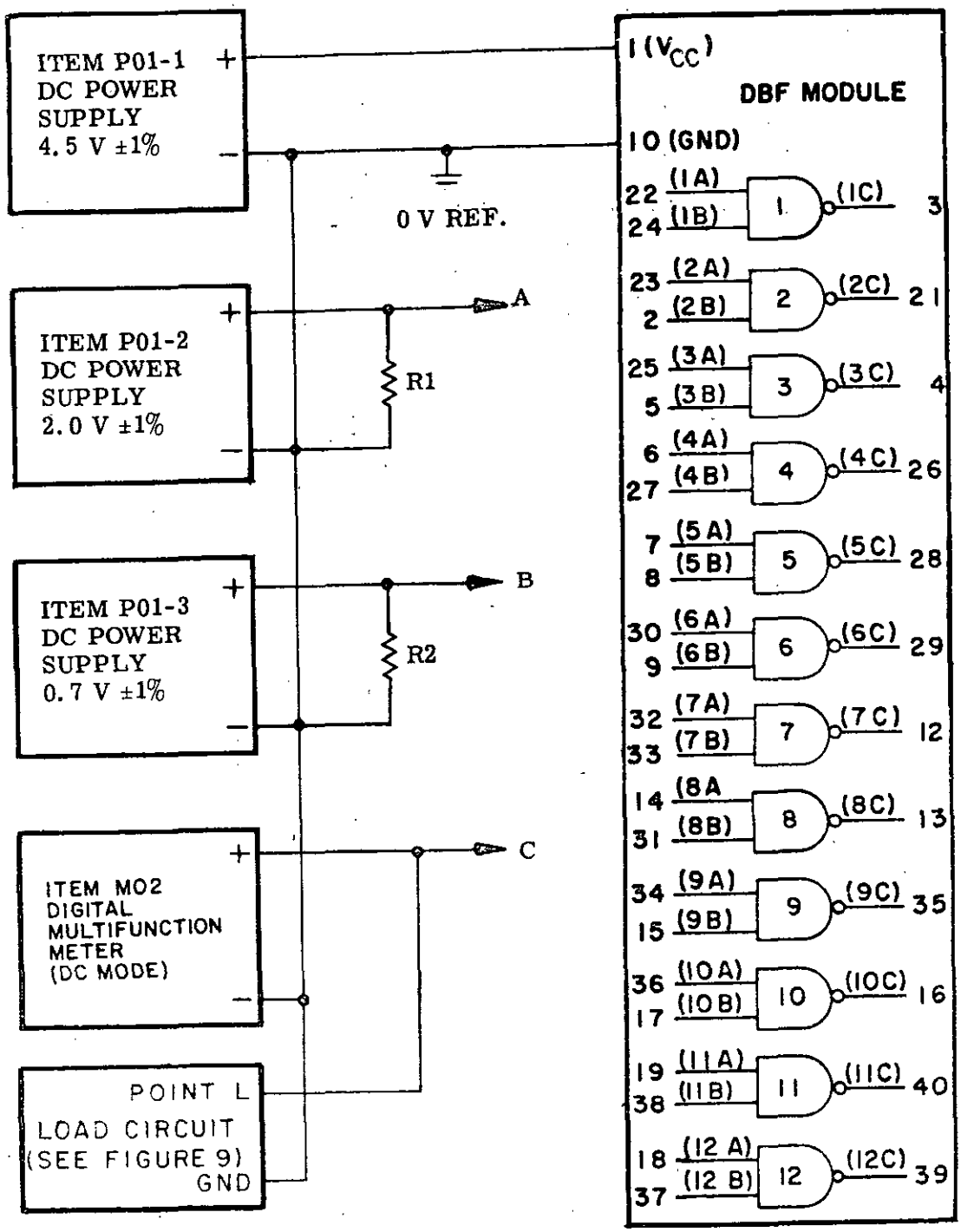
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1C Output voltage, high level	Open	22	3	2.4	4.5	V	8
E2	1C		24	3				
E3	2C		2	21				
E4	2C		23	21				
E5	3C		5	4				
E6	3C		25	4				
E7	4C		6	26				
E8	4C		27	26				
E9	5C		7	28				
E10	5C		8	28				
E11	6C		9	29				
E12	6C		30	29				
E13	7C		32	12				
E14	7C		33	12				
E15	8C		14	13				
E16	8C		31	13				
E17	9C		15	35				
E18	9C		34	35				
E19	10C		17	16				
E20	10C		36	16				
E21	11C		19	40				
E22	11C		38	40				

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TABLE VII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E23	12C Output voltage, high level	Open	18	39	2.4	4.5	V	8
E24	12C Output voltage, high level	Open	37	39	2.4	4.5	V	8

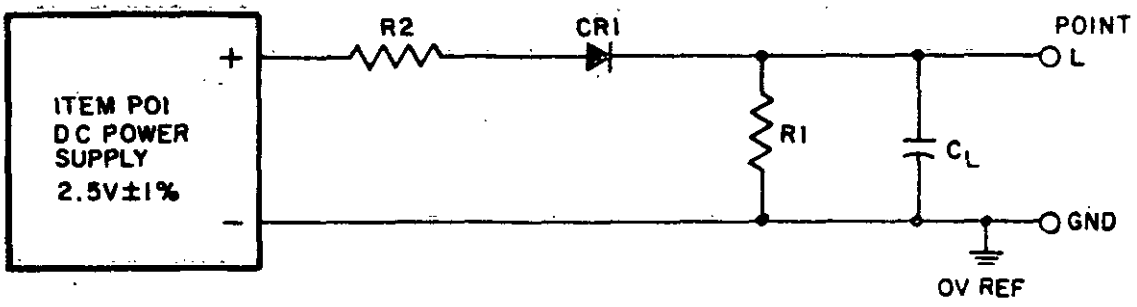
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Since items P01-2 and P01-3 are not used simultaneously, only two item P01 power supplies are required.

FIGURE 8. Output voltage (low-level and high-level) test circuits for tables VI and VII.

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NOTES:
1. CR1 = 1N916 or equivalent
2. R1 = 23.7 kΩ ±1%, 1/8 W (current flow of 100 µA into point L when point L is held at 2.4 V).
3. R2 = select value for current flow of 2.0 mA ±5% out of point L when point L is held at 0.3 V.
4. C_L = 40 ±5 pF (including probe capacitance) for test conditions K1-K12, L1-L12, M1-M12 and N1-N12.
   = 140 ±5 pF (including probe capacitance) for test conditions K13-K24, L13-L24, M13-M24 and N13-N24.
   = 50 ±5 pF (including probe capacitance) for test conditions P1-P12 and R1-R12.
5. C_L is not required for dc electrical tests D and E.
6. The maximum lead length from point "L" to the appropriate module pin shall be less than 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1B Input current, low level	Open	24	Open	22	50	200	µA	10
F2	1A		22		24				
F3	2A		23		2				
F4	2B		2		23				
F5	3A		25		5				
F6	3B		5		25				
F7	4B		27		6				
F8	4A		6		27				
F9	5B		8		7				
F10	5A		7		8				
F11	6A		30		9				
F12	6B		9		30				
F13	7A		32		33				
F14	7B		33		32				
F15	8B		31		14				
F16	8A		14		31				
F17	9A		34		15				
F18	9B		15		34				
F19	10A		36		17				
F20	10B		17		36				
F21	11B		38		19				
F22	11A		19		38				
F23	12A		37		18				
F24	12B		18		37				

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TABLE IX. Input current (high-level) test.

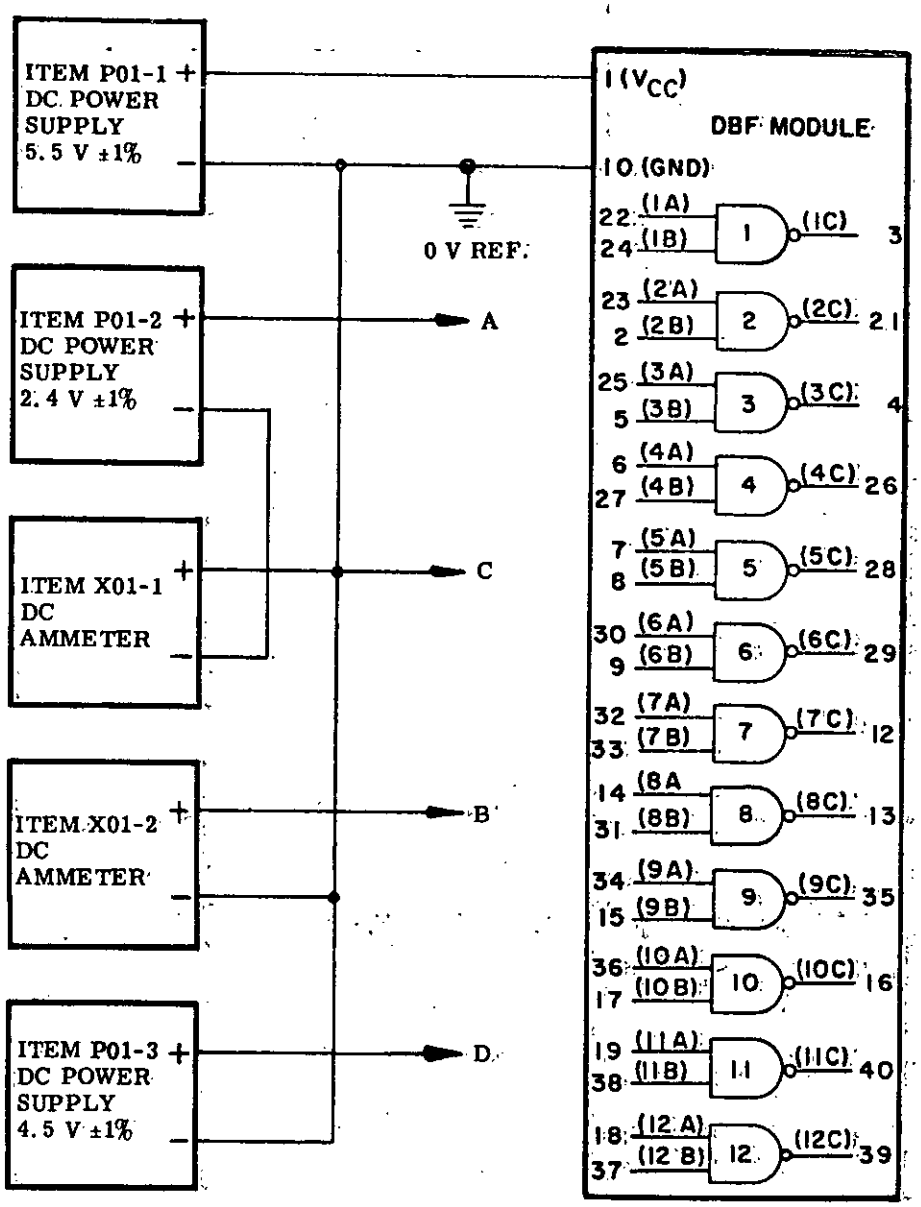
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1B Input current, high level	22	Open	24	Open	0	10	μA	10
G2	1A	24		22					
G3	2A	2		23					
G4	2B	23		2					
G5	3A	5		25					
G6	3B	25		5					
G7	4B	6		27					
G8	4A	27		6					
G9	5B	7		8					
G10	5A	8		7					
G11	6A	9		30					
G12	6B	30		9					
G13	7B	32		33					
G14	7A	33		32					
G15	8B	14		31					
G16	8A	31		14					
G17	9A	15		34					
G18	9B	34		15					
G19	10A	17		36					
G20	10B	36		17					
G21	11B	19		38					
G22	11A	38		19					
G23	12B	18		37					
G24	12A	37		18					

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1C Output short circuit current	Open	3	24, 22	Open	3	15	mA	10
H2	2C		21	23, 2					
H3	3C		4	25, 5					
H4	4C		26	27, 6					
H5	5C		28	8, 7					
H6	6C		29	30, 9					
H7	7C		12	33, 32					
H8	8C		13	31, 14					
H9	9C		35	34, 15					
H10	10C		16	36, 17					
H11	11C		40	38, 19					
H12	12C		39	37, 18					

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since the ammeters are not used simultaneously, only one ammeter, X01, is required.
2. Since items P01-2 and P01-3 are not used simultaneously, only two item P01 power supplies are required.
3. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

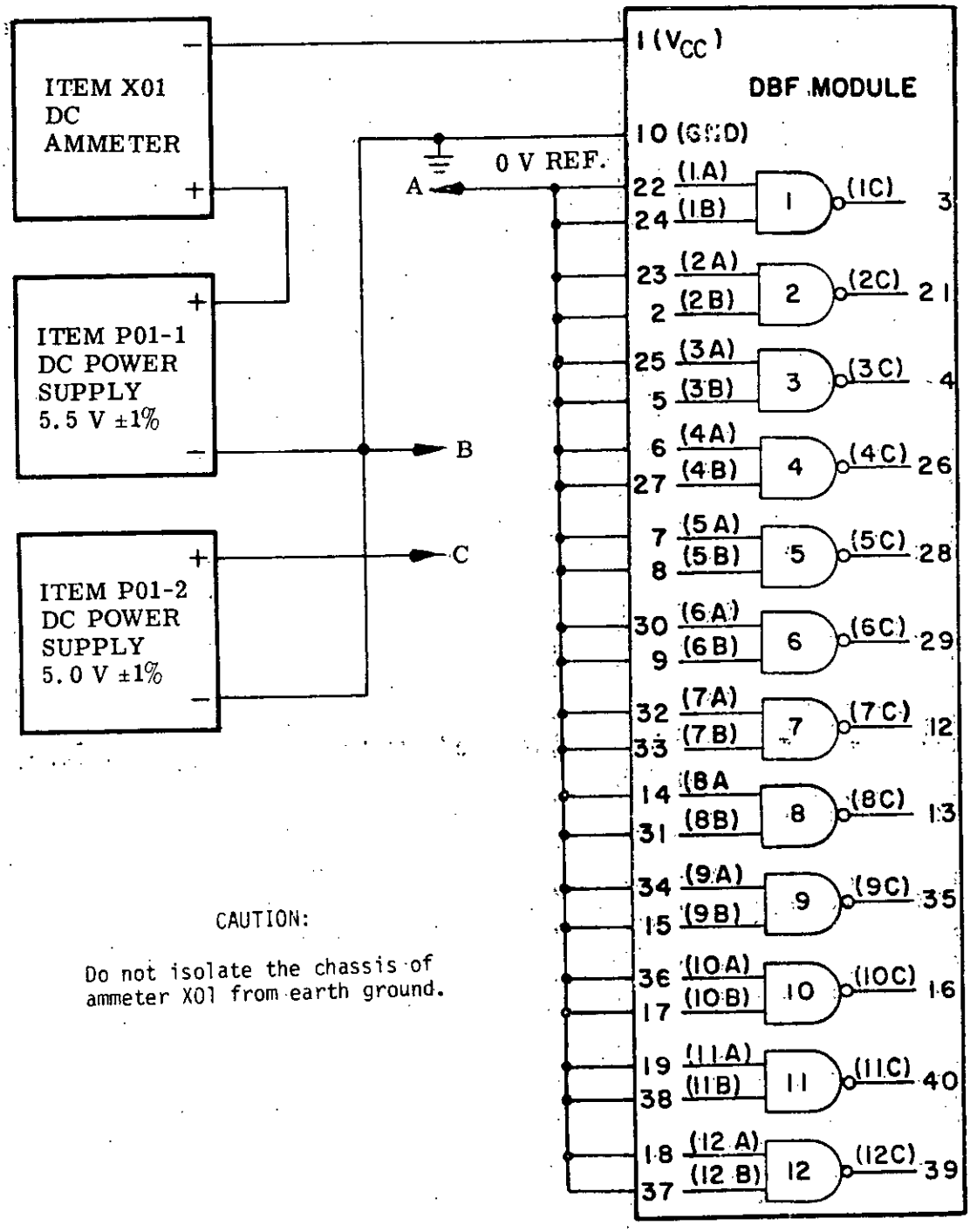
FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Power supply current, low level	C	Open	A	0.5	6.1	mA	11
J2	Power supply current, high level	B	A	Open	0.2	2.4	mA	11

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 11. Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1C Transition time (t_{TLH}), C_L = 40 pF	22	24	3	5	60	ns	12, 13
K2	2C	2	23	21				
K3	3C	5	25	4				
K4	4C	6	27	26				
K5	5C	7	8	28				
K6	6C	9	30	29				
K7	7C	32	33	12				
K8	8C	14	31	13				
K9	9C	15	34	35				
K10	10C	17	36	16				
K11	11C	19	38	40				
K12	12C	18	37	39				
K13	1C Transition time (t_{TLH}), C_L = 140 pF	22	24	3	10	100		
K14	2C	2	23	21				
K15	3C	5	25	4				
K16	4C	6	27	26				
K17	5C	7	8	28				
K18	6C	9	30	29				
K19	7C	32	33	12				
K20	8C	14	31	13				
K21	9C	15	34	35				
K22	10C	17	36	16				
K23	11C	19	38	40				
K24	12C	18	37	39				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1C Transition time (t_{THL}), C_L = 40 pF	22	24	3	2	30	ns	12, 13
L2	2C	2	23	21				
L3	3C	5	25	4				
L4	4C	6	27	26				
L5	5C	7	8	28				
L6	6C	9	30	29				
L7	7C	32	33	12				
L8	8C	14	31	13				
L9	9C	15	34	35				
L10	10C	17	36	16				
L11	11C	19	38	40				
L12	12C	18	37	39				
L13	1C Transition time (t_{THL}) C_L = 140 pF	22	24	3	5	60		
L14	2C	2	23	21				
L15	3C	5	25	4				
L16	4C	6	27	26				
L17	5C	7	8	28				
L13	6C	9	30	29				
L19	7C	32	33	12				
L20	8C	14	31	13				
L21	9C	15	34	35				
L22	10C	17	36	16				
L23	11C	19	38	40				
L24	12C	18	37	39				

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TABLE XIV. Propagation delay time (t_{PLH}) test.

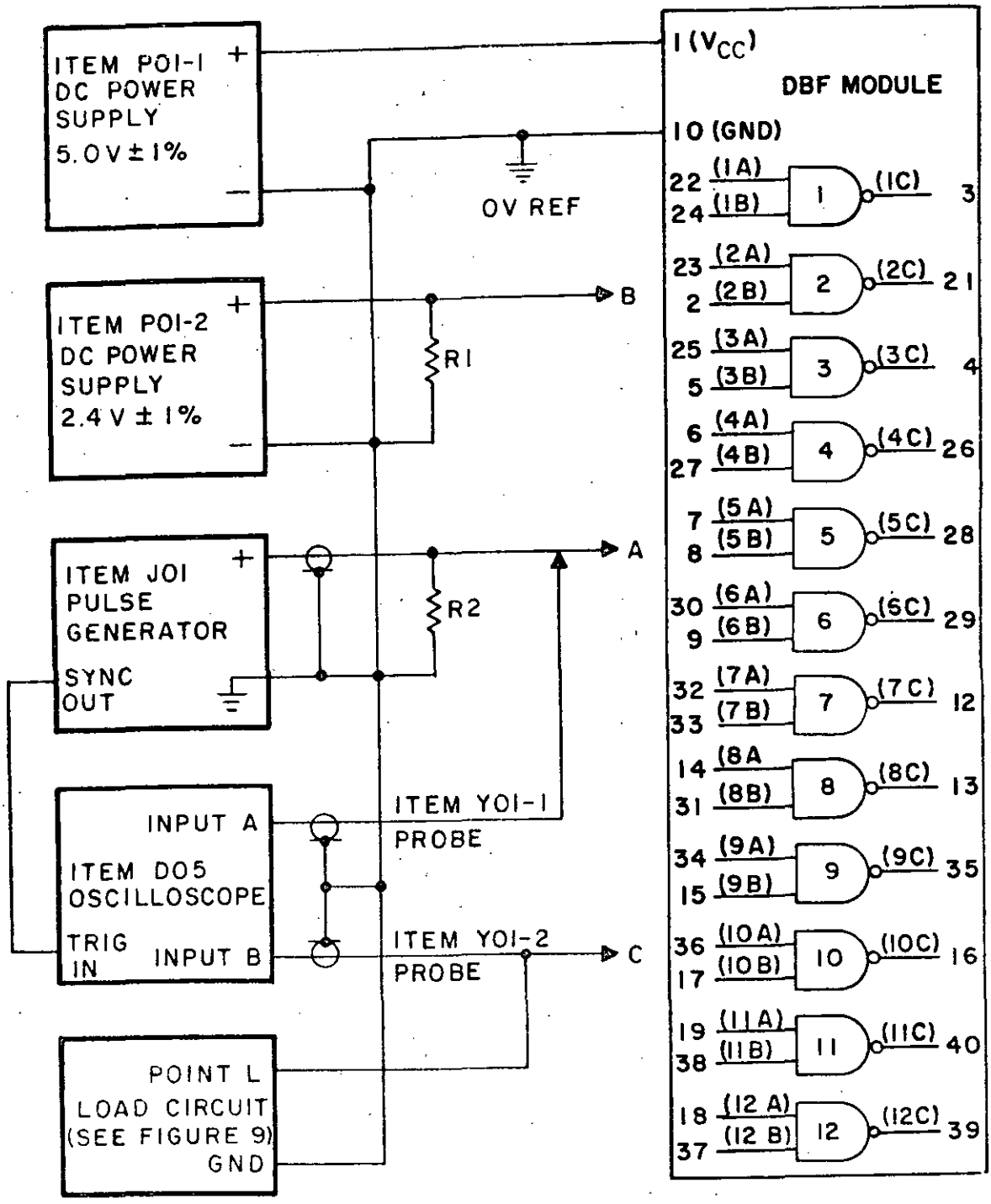
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1C Propagation delay time (t_{PLH}), C_L = 40 pF	22	24	3	10	60	ns	12, 13
M2	2C	2	23	21				
M3	3C	5	25	4				
M4	4C	6	27	26				
M5	5C	7	8	28				
M6	6C	9	30	29				
M7	7C	32	33	12				
M8	8C	14	31	13				
M9	9C	15	34	35				
M10	10C	17	36	16				
M11	11C	19	38	40				
M12	12C	18	37	39				
M13	1C Propagation delay time (t_{PLH}), C_L = 140 pF	22	24	3	10	90		
M14	2C	2	23	21				
M15	3C	5	25	4				
M16	4C	6	27	26				
M17	5C	7	8	28				
M18	6C	9	30	29				
M19	7C	32	33	12				
M20	8C	14	31	13				
M21	9C	15	34	35				
M22	10C	17	36	16				
M23	11C	19	38	40				
M24	12C	18	37	39				

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1C Propagation delay time (t_{PHL}), C_L = 40 pF	22	24	3	10	82	ns	12, 13
N2	2C	2	23	21				
N3	3C	5	25	4				
N4	4C	6	27	26				
N5	5C	7	8	28				
N6	6C	9	30	29				
N7	7C	32	33	12				
N8	8C	14	31	13				
N9	9C	15	34	35				
N10	10C	17	36	16				
N11	11C	19	38	40				
N12	12C	18	37	39				
N13	1C Propagation delay time (t_{PHL}), C_L = 140 pF	22	24	3	10	96		
N14	2C	2	23	21				
N15	3C	5	25	4				
N16	4C	6	27	26				
N17	5C	7	8	28				
N18	6C	9	30	29				
N19	7C	32	33	12				
N20	8C	14	31	13				
N21	9C	15	34	35				
N22	10C	17	36	16				
N23	11C	19	38	40				
N24	12C	18	37	39				

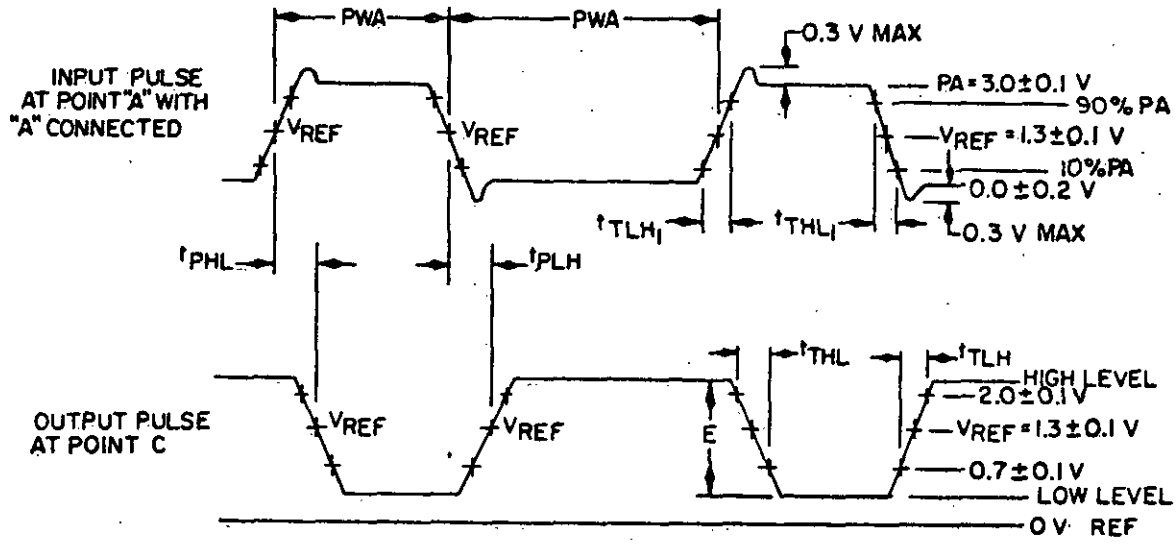
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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII, XIV, and XV.

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NOTES:
1. Input pulse parameters.
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
2. Before making the measurements verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.
3. All voltages are referenced to ground.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII, XIV, and XV.

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TABLE XVI. Output crosstalk (low-level) test.

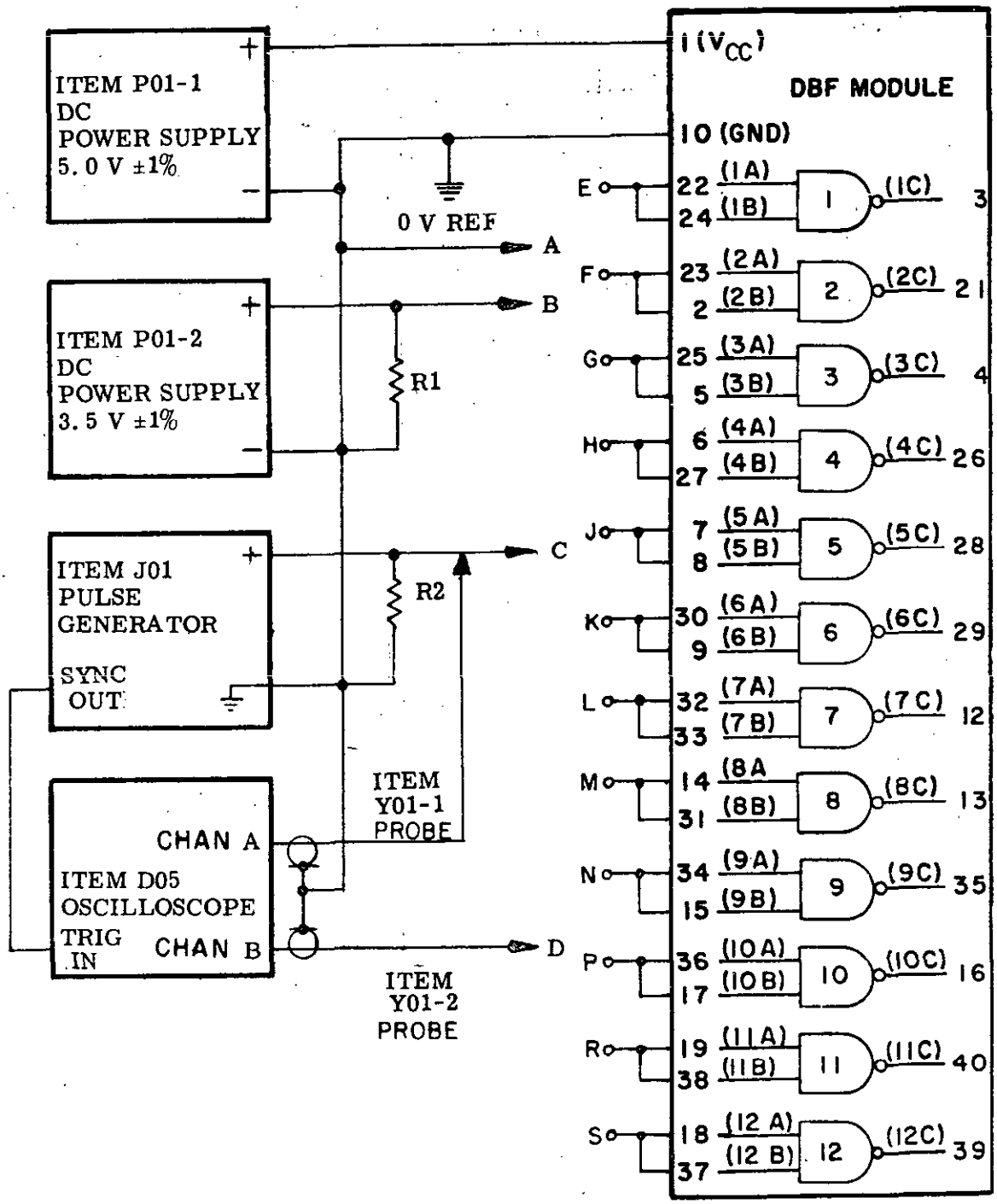
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1C Output crosstalk, low level	Open	E	F,G,H,J,K, L,M,N,P,R,S	3	0	400	mV	14, 15
P2	2C		F	E,G,H,J,K, L,M,N,P,R,S	21				
P3	3C		G	E,F,H,J,K, L,M,N,P,R,S	4				
P4	4C		H	E,F,G,J,K, L,M,N,P,R,S	26				
P5	5C		J	E,F,G,H,K, L,M,N,P,R,S	28				
P6	6C		K	E,F,G,H,J, L,M,N,P,R,S	29				
P7	7C		L	E,F,G,H,J, K,M,N,P,R,S	12				
P8	8C		M	E,F,G,H,J, K,L,N,P,R,S	13				
P9	9C		N	E,F,G,H,J, K,L,M,P,R,S	35				
P10	10C		P	E,F,G,H,J, K,L,M,N,R,S	16				
P11	11C		R	E,F,G,H,J, K,L,M,N,P,S	40				
P12	12C		S	E,F,G,H,J, K,L,M,N,P,R	39				

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1C Output crosstalk, high level	E	Open	F,G,H,J,K,L, M,N,P,R,S	3	0	400	mV	14, 15
R2	2C	F		E,G,H,J,K,L, M,N,P,R,S	21				
R3	3C	G		E,F,H,J,K,L, M,N,P,R,S	4				
R4	4C	H		E,F,G,J,K,L, M,N,P,R,S	26				
R5	5C	J		E,F,G,H,K,L, M,N,P,R,S	28				
R6	6C	K		E,F,G,H,J,L, M,N,P,R,S	29				
R7	7C	L		E,F,G,H,J,K, M,N,P,R,S	12				
R8	8C	M		E,F,G,H,J,K, L,N,P,R,S	13				
R9	9C	N		E,F,G,H,J,K, L,M,P,R,S	35				
R10	10C	P		E,F,G,H,J,K, L,M,N,R,S	16				
R11	11C	R		E,F,G,H,J,K, L,M,N,P,S	40				
R12	12C	S		E,F,G,H,J,K, L,M,N,P,R	39				

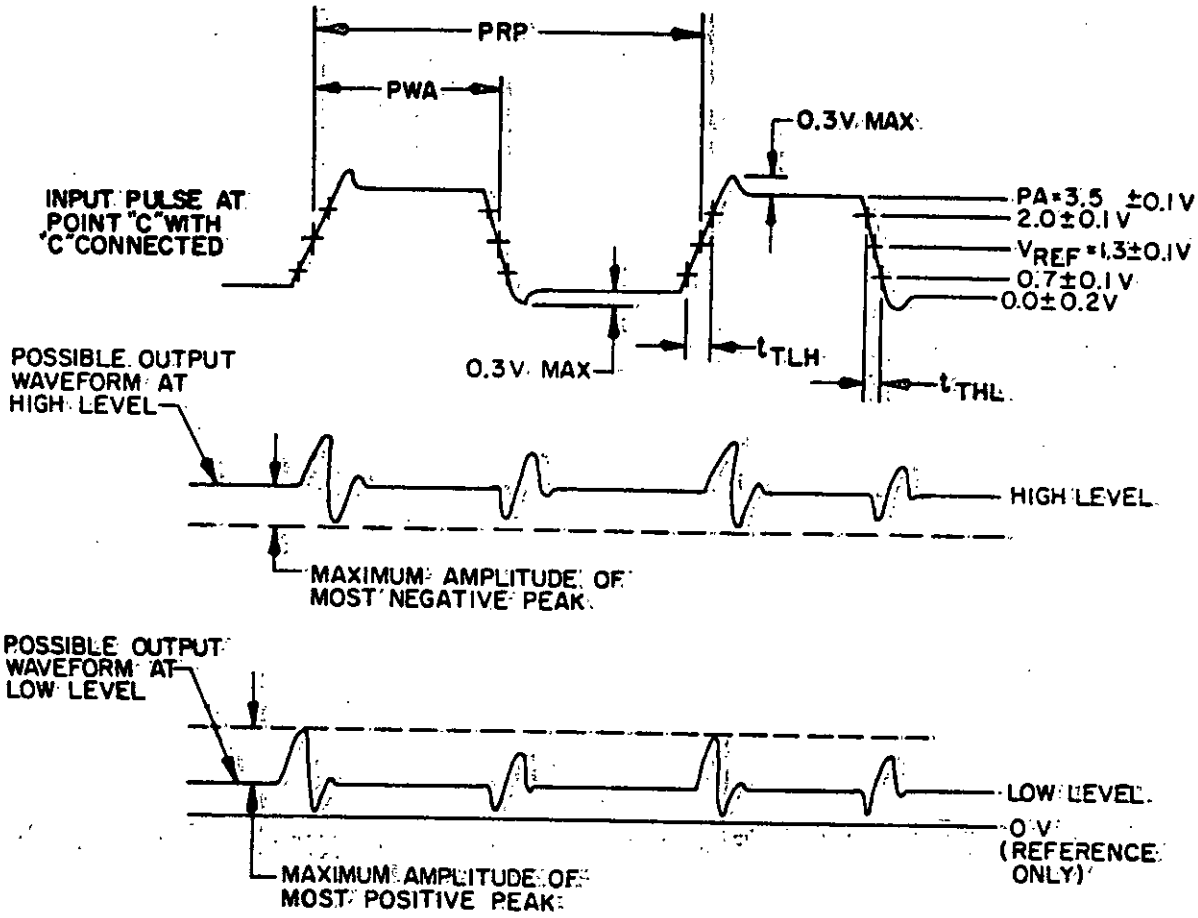
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all output with the load specified on figure 9.

FIGURE 14. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTE:
Input pulse parameters:
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all inputs).
PWA = 200 ±20 ns.
PRP = 1,000 ±100 ns.

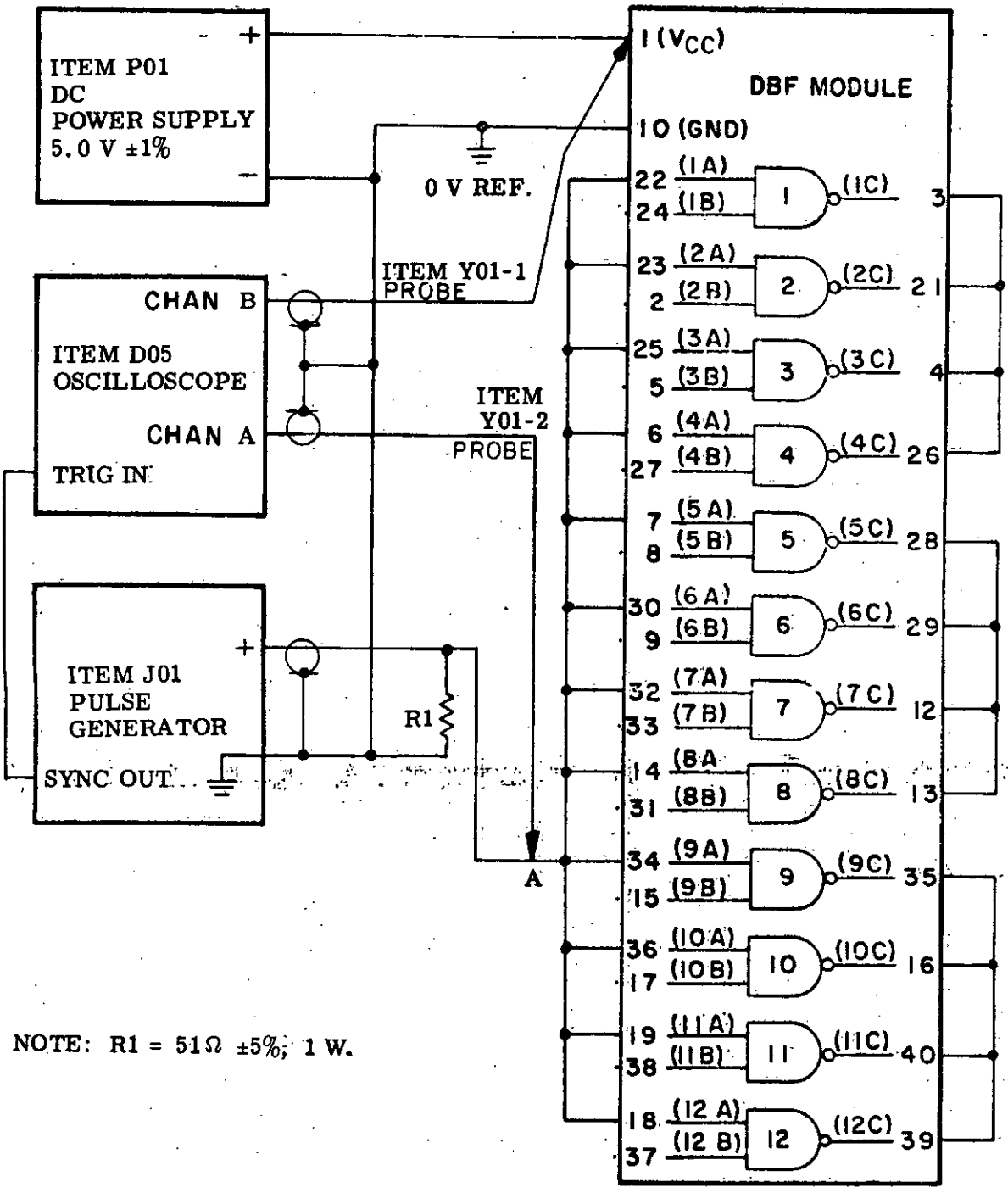
FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	300	mV	16, 17

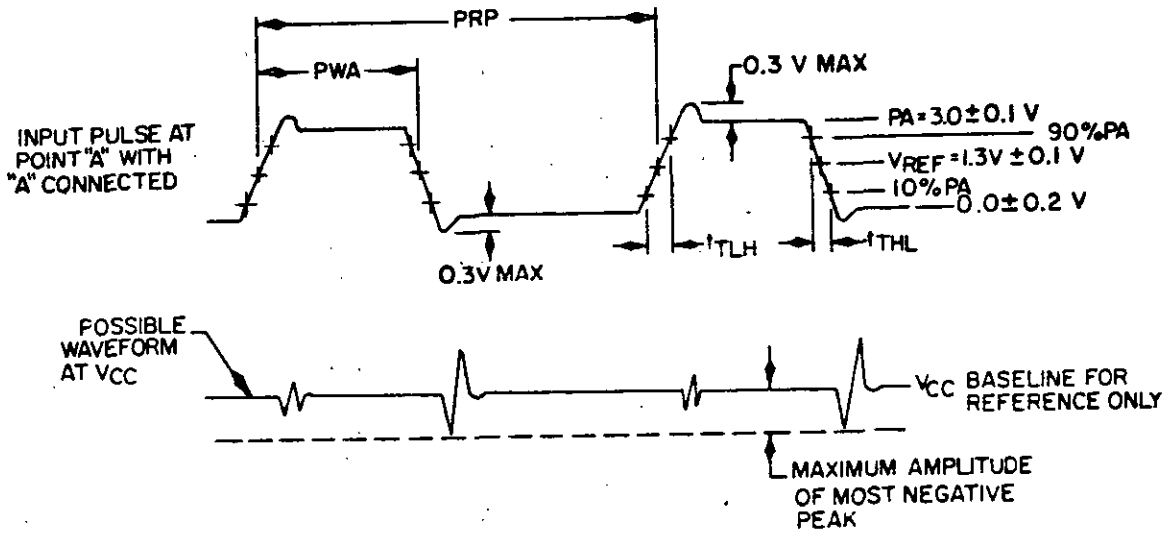
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NOTE: R1 = 51Ω ±5%; 1 W.

FIGURE 16. Power supply transient amplitude test circuit for table XVIII.

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NOTE:
Input pulse parameters:
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.
PWA = 200 ±20 ns.
PRP = 1,000 ns ±100 ns.

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX. Power supply filter capacitance test.

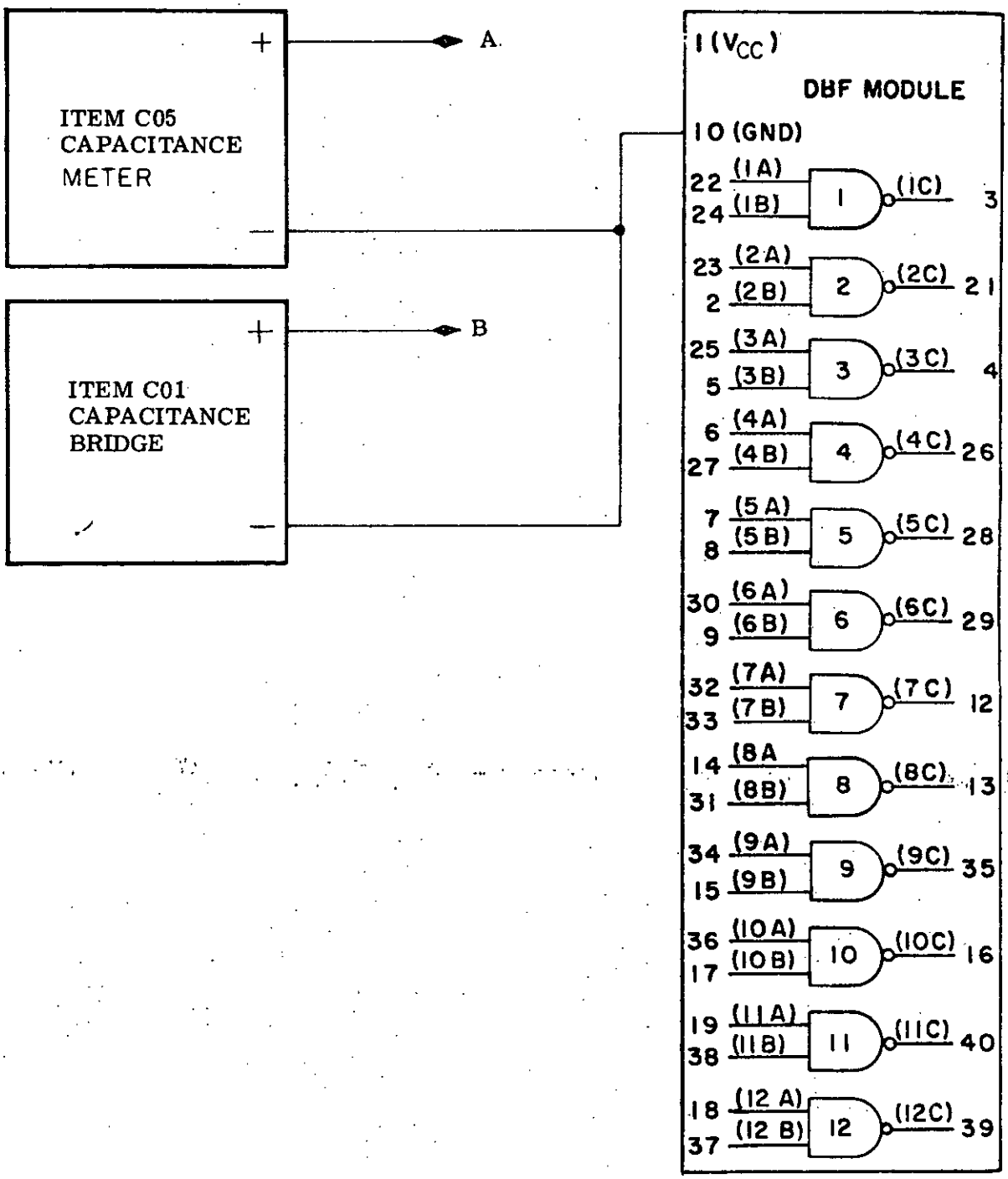
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	1	1.65	2.75	µF	18

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	1A Input capacitance	22	Open	2	15	pF	18
U2	1B	24					
U3	2B	2					
U4	2A	23					
U5	3B	5					
U6	3A	25					
U7	4A	6					
U8	4B	27					
U9	5A	7					
U10	5B	8					
U11	6B	9					
U12	6A	30					
U13	7A	32					
U14	7B	33					
U15	8A	14					
U16	8B	31					
U17	9B	15					
U18	9A	34					
U19	10B	17					
U20	10A	36					
U21	11A	19					
U22	11B	38					
U23	12A	18					
U24	12B	37					

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V max.

FIGURE 18. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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TABLE XXI. Output impedance (power off) check test.

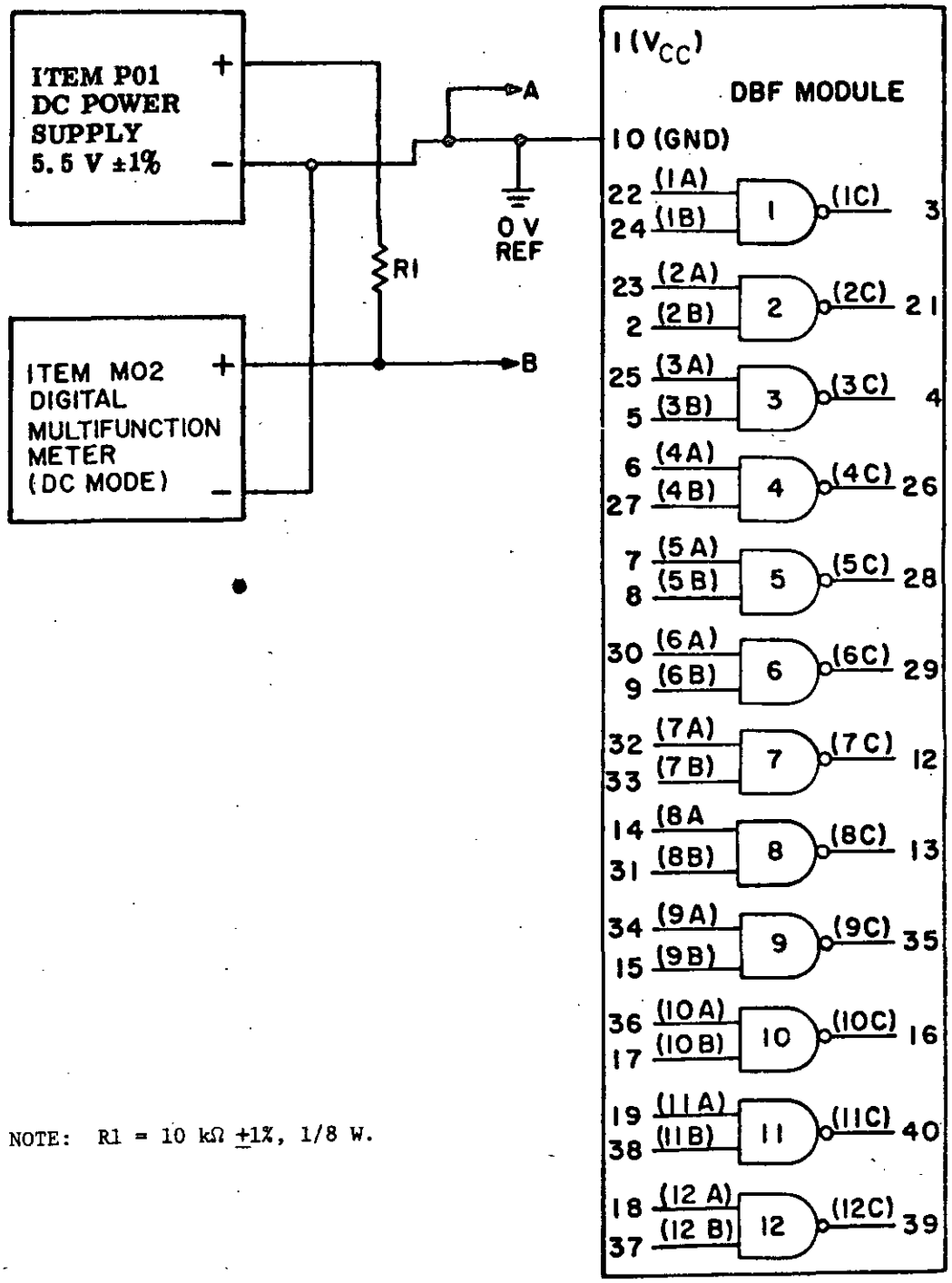
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	1C Output impedance, power off, check	Open	3	3.50	5.55	V	19
V2	2C		21				
V3	3C		4				
V4	4C		26				
V5	5C		28				
V6	6C		29				
V7	7C		12				
V8	8C		13				
V9	9C		35				
V10	10C		16				
V11	11C		40				
V12	12C		39				
V13	1C	1	3				
V14	2C		21				
V15	3C		4				
V16	4C		26				
V17	5C		28				
V18	6C		29				

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TABLE XXI. Output impedance (power off) check test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V19	7C Output impedance, power off, check	1	12	3.50	5.55	V	19
V20	8C		13				
V21	9C		35				
V22	10C		16				
V23	11C		40				
V24	12C		39				

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NOTE: R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 19. Output impedance (power off) check test circuit for table XXI.

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